

Mata

J. Mann

**A comprehensive guide to
the language**

Written in English

April 2026

Mata

0 Introduction

Ēŷ_(Ehh-ih-ya) | **Hello**

Welcome to this comprehensive guide to my language known as *Mata*.

This is my first conlang, made to revolve around mathematical notation, in particular set notation. The core concept to know is that almost everything can be described as a set, containing all of the information about said thing. For example, a person can be stated to be a set (i.e. $\mathbb{P}$) that contains all of the adjectives describing them. We can then make statements about that set (i.e. " $\mathbb{P} \ni \text{stupid}$ " *aka* the person is stupid, however "stupid" wouldn't just be in English).

Mata

1 Lexicon

1.1 Alphabet

Using the IPA for the sounds

Symbol	Sound	Symbol	Sound	Symbol	Sound	Symbol	Sound	Symbol	Sound
Aa	æ	Gg	g	Mm	m	Ss	s	Yy	j
Bb	b	Hh	h	Nn	n	Tt	t	Zz	z
Cc	tʃ	Ii	ɪ	Oo	ʊ	Uu	ʊ		
Dd	d	Jj	ʒ	Pp	p	Vv	v		
Ee	ɛ	Kk	k	Qq	θ	Ww	w		
Ff	f	Ll	l	Rr	r	Xx	ʃ		

For other sounds, the symbol (,) can be used to modify certain symbols. For vowels it converts them into their long forms
(Written in L^AT_EX as ”_\prime”)

List of alternate sounds:

Symbol	Sound
<i>a,</i>	eɪ
<i>e,</i>	i:
<i>i,</i>	aɪ
<i>o,</i>	oʊ
<i>u,</i>	u:
<i>q,</i>	ð

”*q,*” is very uncommon but does exist, primarily for words taken from other languages being written in Mata’s alphabet.
For example, ”The” would be written as ”*q,*^h”. The other symbols in this will be touched on shortly.

Mata very rarely uses the normal symbols for vowels outside of the beginning of words, or pluralisation. Instead, Mata uses diacritics to denote when a vowel follows a consonant.

A table of the diacritic marks follows:

Symbol	Equivalent vowel	L ^A T _E X
ô	a	\hat{...}
ö	e	\ddot{...}
õ	i	\tilde{...}
ó	o	\dot{...}
ö	u	\breve{...}

Additionally there are two other diacritics other than for vowels.

Symbol	Sound	Symbol	Sound
o ^h	Aspirated	Ŝ	Long S, short a
ō / o	Long Sound	ŝ	Short S, long a

Mata

1.2 Special Characters

Mata uses more than a basic alphabet. There are a few special characters with dedicated meanings. All special characters from now on will have their sounds written in the Mata alphabet.

Symbol	L ^A T _E X	Sound	Meaning
$\mathcal{E}$	<code>\mathcal{E}</code>	<i>e</i> ,	The speaker (I)
$\mathfrak{U}$	<code>\mho</code>	<i>û</i>	The listener (you singular)
$\emptyset$	<code>\emptyset</code>	<i>h</i>	0 / nothing
ξ	<code>\xi</code>	<i>a^h</i>	Everything
σ	<code>\sigma</code>	<i>š</i> , <i>g</i>	Level of significance

1.3 Operators

Operators are special characters which describe properties or modifications to lexical objects. In sentences, they are particles that fit between objects.

Symbol	L ^A T _E X	Sound	Meaning
$+$	<code>+</code>	<i>płs</i>	Addition
$-$	<code>-</code>	<i>ññs</i>	Subtraction
$\times$	<code>\times</code>	<i>mlt</i>	Multiplication
$\div$	<code>\div</code>	<i>đv̇j</i>	Division
$\ni / \in$	<code>\ni / \in</code>	<i>k̇</i> or <i>k̇</i> ,	Contains / Is contained in
$\not\ni / \notin$	<code>\not\ni / \notin</code>	<i>ṅ</i> $\ni$ or <i>ṅ</i> , $\ni$	Does not contain / is not contained in
$\circ$	<code>\circ</code>	<i>k</i> (short)	Used to access an object's child
$\cdot$	<code>\cdot</code>	<i>also k</i> (short)	Decimal place
$\neg$	<code>\neg</code> (often on keyboards)	<i>ṅ</i> or <i>ṅ</i> ,	Not, for saying something is false
$\dashv$	<code>\dashv</code>	<i>ṁ</i> or <i>ṁ</i> ,	Not, for inverting
$\vdash$	<code>\vdash</code>	<i>oyl</i>	Was true and is still true.
$\updownarrow$	<code>\updownarrow</code>	<i>vr</i>	Vertical
$\leftrightarrow$	<code>\leftrightarrow</code>	<i>hṙ</i>	Horizontal
$=$	<code>=</code>	<i>a^h</i>	is/equals
$\neq$	<code>\neq</code>	<i>na^h</i>	is not/not equal
$\overset{*}{=}$	<code>\overset{*}{=}</code>	<i>oŕ</i>	Requires/depends on
$\supset / \subset$	<code>\supset / \subset</code>	<i>ṡ</i> or <i>ṡ</i> ,	Is a subset of / contains the set... (plural of $\in$)
$\not\supset / \not\subset$	<code>\not\supset / \not\subset</code>	<i>ṅ</i> $\supset$ or <i>ṅ</i> , $\supset$	negative of subset/superset
Next page...			

Mata

Symbol	L ^A T _E X	Sound	Meaning
?	?	<i>v^h</i>	Convert an operator to a question
{ / }	\{ / \}	<i>î</i>	Open/Close a set. "}" does not have to be pronounced at the end of a sentence.
		<i>Ǧ</i>	And, in a set
∩	\cap	<i>Ǧ_iy</i>	And, between sets (intersection)
∪	\cup	<i>Ǧ_y</i>	Or, between sets (union)
∴	\therefore	<i>ǦǦ_i</i>	Therefore
∵	\because	<i>ǦǦ_i</i>	Because
◊	\diamond	<i>Ǧ</i>	Is possible (can)
⊞	\boxplus	<i>ǦpǦ</i>	However/But (Despite this)
∧	\land	<i>Ǧ</i>	And, in logic
∨	\vee	<i>Ǧ</i>	Or, in logic
:	\colon	No sound	Used optionally to separate adjacent words as Mata does not use spaces

1.4 Numbers

Mata, like other western languages, uses base 10 and the same numerals as English (Hindu-Arabic numerals if i remember correctly). The names for numbers are as follows:

Numeral	Word	Numeral	Word	Numeral	Word
0	<i>na</i>	10	<i>ǧ_in</i>	100	<i>ǧ_it</i>
1	<i>ak</i>	11	<i>ǧ_inak</i>	200	<i>ǧ_it</i>
2	<i>el</i>	20	<i>ǧ_in</i>	300	<i>ǧ_it</i>
3	<i>dǦ</i>	30	<i>ǧ_in</i>	1000	<i>ǧ_iǧ</i>
4	<i>fǧ</i>	40	<i>ǧ_in</i>	2000	<i>ǧ_iǧ</i>
5	<i>ǧn</i>	50	<i>ǧ_in</i>	3000	<i>ǧ_iǧ</i>
6	<i>ǧn</i>	60	<i>ǧ_in</i>		
7	<i>pǦ</i>	70	<i>pǦ_in</i>		
8	<i>uv</i>	80	<i>ǧ_in</i>		
9	<i>ǧc</i>	90	<i>ǧ_in</i>		

If you wish to say a number with order above 3, you can use scientific notation with the *p* particle (i.e 2.5 million = 2.5 × 10⁶ = *elkǧnǧpǧn*).

Numbers can be used for emphasis like an adjective (i.e. "very good" would be like saying "10 good"/"*ǧ_in : ǧǧl*"). The ordinal variants are as follows:

Number	1	2	3	4	5	6	7	8	9	10	11	20	30	100
Word	<i>ak_i</i>	<i>ǧā</i>	<i>dǦ</i>	<i>f_i</i>	<i>ǧ_it</i>	<i>ǧw</i>	<i>pǦ_i</i>	<i>ǧl</i>	<i>ǧĉ</i>	<i>ak_in</i>	<i>ak_inak</i>	<i>ǧ_in</i>	<i>ǧd_in</i>	<i>ak_it</i>

Mata

2 Words

Time for an overview of some of the words in *Mata*. These lists are not extensive and at somepoint I will create a dictionary for them.

2.1 Verbs

Verbs is not the best word to describe these words, they more represent the concept of the action rather than the action itself. This will be more relevant later. Here is a list of some of the more common *Verbs* (or some of the first few I created) with rough English translations:

English	Mata	English	Mata	English	Mata	English	Mata	English	Mata
To do	$\acute{d}$	To say/speak	$\acute{S}g$	To play	$\acute{s}nd$	To hate	$\acute{d}\ddot{r}$	To fish	$\acute{s}\hat{m}$
To read	$k\ddot{o}c$	To go	$\check{m}ln$	To consume	$v\acute{l},s$	To get	$\acute{r}\hat{r}$	To know	$o\acute{y}lr$
To want	$\acute{y}tx$	To see	$b\hat{y}n$	To sex	$\check{l}\hat{f}$	To assume/ think/ believe	$\acute{d}rs$	To milk	$\check{m}lj$
To be going to/intend to (will)	$\check{q}\hat{v}t$	To assert (suggest is true)	$\check{k}l\hat{t}$	To die/kill	$o\acute{l}k$				

Mata supports compound *verbs* as well:

English	Mata	English	Mata	English	Mata	English	Mata	English	Mata
To sleep	$n\acute{y}c\check{r}ln$	To drive	$\hat{m}\tilde{x}n\check{m}ln$	To love	$\neg \acute{d}\ddot{r}$	To hunger/ thirst	$v\acute{l},s\acute{y}tx$		

In order to describe the past or future tense of a *verb*, you can either

- Access the past or future child of the *verb* (i.e. $\mathcal{E} \circ \acute{d} \circ \tilde{u}p$ is the set of things that I, the speaker, have done)
- Apply the past or future as a function, less uses (i.e. $\tilde{u}p(\hat{s}\hat{m})$ is the state of having fished, The sentence "he is going to have fished" would be $\tilde{G} \circ \acute{d} \circ k\check{l} \ni \tilde{u}p(\hat{s}\hat{m}))$

Here are a few example sentences:

<i>James says "Hello"</i>	$D\hat{j},mz \circ \hat{S}g \ni "$	$\check{H}\acute{l}_i,"$	(James'k sag koh Hello)
<i>James is saying "Hello"</i>	$D\hat{j},mz \circ \acute{d} \ni \hat{S}g("$	$\check{H}\acute{l}_i,")$	(James'k doh koh sag Hello)
<i>James is doing nothing</i>	$D\hat{j},mz \circ \acute{d} = \emptyset$		(James'k doh ah na)
<i>I hate you</i>	$\mathcal{E} \circ \acute{d}\ddot{r} \ni \mathcal{U}$		(Ee'k dareh koh wa)
<i>I hate you a lot</i>	$\mathcal{E} \circ \acute{d}\ddot{r} : \check{k},n \ni \mathcal{U}$		(Ee'k dareh keen koh wa)
<i>I really really love you</i>	$\mathcal{E} \circ \neg \acute{d}\ddot{r} : \check{k},\hat{y} \ni \mathcal{U}$		(Ee'k modareh keeya koh wa)
<i>I saw you</i>	$\mathcal{E} \circ b\hat{y}n \circ \tilde{u}p \ni \mathcal{U}$		(Ee'k byan k'u-ip koh wa)
<i>I saw you reading</i>	$\mathcal{E} \circ b\hat{y}n \circ \tilde{u}p \ni \mathcal{U} \circ \acute{d}(k\ddot{o}c)$		(Ee'k byan k'u-ip koh wa'k doh(ko-ich))
<i>James can't do everything</i>	$D\hat{j},mz \circ \diamond \acute{d} \neq \xi$		(James k'padoh na ah)
<i>I want to go to sleep</i>	$\mathcal{E} \circ \acute{y}tx \ni n\acute{y}c\check{r}ln$		(Ee'k yetsh koh nyoch-muln)

Mata

2.2 Nouns and Pronouns

2.2.1 Pronouns

Mata supports most of the same pronouns as *English* but remember that objects like pronouns are sets which can be combined together to describe different groups using the "∩" and "∪" symbols.

English	Mata	English	Mata	English	Mata	English	Mata	English	Mata
The Speaker, I	$\mathcal{E}$	The Listener, You[sing]	$\mathcal{U}$	They/It*	$\tilde{g}$	You[plur] [†]	$p\hat{y}$	They [plur] [‡]	$g\hat{y}$
This/These	$\mathcal{E}\hat{c}$	That/Those	$\mathcal{U}\hat{c}$						

*Any physical object or person. Cannot be used for anything abstract, except for personification

[†]Any group of people including the listener and not including the speaker

[‡]Any group of people including neither the listener nor the speaker

2.2.2 Nouns

English	Mata	English	Mata	English	Mata	English	Mata	English	Mata
Past	$\tilde{u}p$	Future	$k\check{l}$	Name	$\tilde{n}ym$	Night	$n\acute{y}c$	Apple	$a\check{p}l$
Car	$\acute{m}\tilde{x}n$	Head	$\hat{l}gk$	Mum	$\acute{m}\check{m},$	Dad	$\hat{p}\check{p},$	Child	$\tilde{k}\hat{l}n$

Mata

2.3 Adjectives

Adjectives are primarily made of root words and modifier particles. For example, the word for short is equivalent to 'not vertical length'. Compound words can also be made by combining other words. For example, enjoyment is equivalent to 'doing-happiness'.

Roots:

English	Mata	English	Mata	English	Mata	English	Mata	English	Mata
Length	$\hat{k}\hat{r}$	Happiness	$\hat{t}/\hat{r}l$	Good	$\hat{T}\hat{n}l$	Bad	$\check{H}/\check{n}l$	Empty	$\hat{v}n$
Area	$E\hat{r}$	Volume	$\hat{t}\hat{r}$	Male	$\hat{k}l$	Female	$\hat{v}l$		

Even though the words for length, area, and volume describe the concepts, describing an object as having the concept as a quality suggests that it has a large amount. So if I want to say that an object is long I can just say that it has length, and if i want to say that it's short I can say it has not-length. To convert these concepts into physical objects, the relevant affix would be needed. Therefore, *An Area* would be $E\hat{r}\hat{y}n$.

Modifiers:

These are all operators from before, however I will still reiterate the relevant ones here.

English	Mata	English	Mata	English	Mata	English	Mata	English	Mata
Not	$\neg$	Vertical	$\updownarrow$	Horizontal	$\leftrightarrow$				

Mata

2.4 Interjections

2.4.1 Frequently Used

English	Mata	English	Mata	English	Mata	English	Mata	English	Mata
Yes	<i>ŧ</i>	No	<i>ḥ,</i>	Thanks	<i>škr̥l</i>	Hello	<i>Ěŷ</i>	Excuse me	<i>iŵn̄</i>

2.4.2 Expletives

English	Mata	English	Mata	English	Mata	English	Mata	English	Mata
F*ck/ Holy Sh*t	<i>ŷṿḷṛ</i>	B*tch/ Idiot	<i>ḳṛṇṭ</i>						